

Davis RF Encoding Guide - Vantage Pro2 / Pro2 Plus EU - 868 MHz

Edition 1.0 - 31 August 2026

Project: ESP32 Davis Weather Gateway

This document summarizes the RF behaviour implemented by ESP32 Davis Weather Gateway. It is an independent interoperability/reverse-engineering note and **not an official Davis Instruments specification**. Several conversions still require validation on real hardware.

Architecture

```
Davis ISS EU
| 868 MHz / 2-FSK / FHSS
v
SX1276 / RFM95
| SPI
v
ESP32 / LILYGO
|-- Davis decoder
|-- local BME280 pressure
|-- Web UI / diagnostics
`-- optional HTTP upload
```

The project is Davis-only; Oregon Scientific and Technoline decoders are not part of this variant.

RF profile

Parameter	Implemented value
Band	868 MHz EU
Modulation	2-FSK
Bit rate	19.2 kbps
RX deviation	4.8 kHz
Shaping	Gaussian BT=0.5
RX bandwidth	25 kHz
AFC bandwidth	50 kHz
Preamble	4 x 0xAA
Sync	0xCB 0x89
Payload	fixed 10 bytes
CRC	CRC16-CCITT in firmware

The SX1276 hardware CRC is disabled because the Davis frame carries its own CRC.

European hop set

Hop	FRF	Frequency
1	D9 04 45	868.066711 MHz
2	D9 13 04	868.297119 MHz
3	D9 21 C2	868.527466 MHz

Hop	FRF	Frequency
4	D9 0B A4	868.181885 MHz
5	D9 1A 63	868.412292 MHz

The firmware uses an approximately 2555 ms nominal packet interval. After a Davis-shaped packet it advances to the next hop; after prolonged misses it returns to acquisition. This synchronization algorithm is a gateway implementation choice, not a guaranteed reproduction of Davis console internals.

10-byte frame

RF bytes are normalized by bit reversal before field/CRC interpretation.

Byte	Current gateway use
0	packet type + battery flag + transmitter ID
1	wind speed
2	wind direction
3	type-specific payload
4	type-specific payload
5	data/reserved, included in primary CRC
6	CRC MSB
7	CRC LSB
8	additional/retransmit field, not validated in current alpha
9	additional/retransmit field, not validated in current alpha

Byte 0 is treated as TTTT B III: high nibble packet type, bit 3 battery flag, bits 2..0 transmitter ID 0..7. The UI displays IDs 1..8 and supports 0 as auto-lock.

Packet types currently decoded

Type	Name	Measurement
0x4	UV	UV index
0x5	RAIN_RATE	seconds since last rain tip / rain rate
0x6	SOLAR	solar radiation
0x8	TEMP	outside temperature
0x9	WIND_GUST	10-minute gust
0xA	HUMIDITY	outside humidity
0xE	RAIN	rain tip counter

Other valid packet types may be logged as OTHER without updating a sensor value.

CRC16-CCITT

The gateway calculates CRC over normalized bytes 0..5 with polynomial 0x1021 and initial value 0, then compares the result with bytes 6..7 interpreted MSB first.

Current field conversions

Wind speed:

```
wind_kmh = byte1 * 1.609344
```

Direction:

```
wind_dir_deg = 9.0 + byte2 * (342.0 / 255.0)
```

This direction mapping must be checked against a real ISS, especially around the 0/360 degree wrap.

Temperature (0x8):

```
raw = ((byte3 << 8) | byte4) >> 4
temp_f = raw / 10.0
temp_c = (temp_f - 32.0) * 5.0 / 9.0
```

Humidity (0xA):

```
raw = ((byte4 >> 4) << 8) | byte3
humidity_pct = raw / 10.0
```

Gust (0x9):

```
gust_kmh = byte3 * 1.609344
```

Solar radiation (0x6):

```
raw10 = byte3 * 4 + (byte4 >> 6)
solar_wm2 = raw10
```

UV (0x4):

```
raw = ((byte3 << 8) | byte4) >> 4
uv = (raw - 4.0) / 200.0
```

The UV conversion explicitly requires field comparison against a Davis reference.

Rain counter (0xE) uses the low 7 bits of byte 3 as a modulo-128 counter. The configurable rain-tip size converts counter deltas into millimetres. Large anomalous deltas are rejected.

Rain rate (0x5) reconstructs seconds since the last tip from byte 3 and the high nibble of byte 4, then calculates:

```
rain_rate_mm_h = rain_mm_per_tip * 3600 / seconds_since_tip
```

Atmospheric pressure

The architecture does not derive pressure from the ISS RF payload. A local BME280 is used and reduced to sea level using the configured installation altitude:

```
P0 = Pstation / (1 - altitude_m / 44330) ^ 5.255
```

FHSS synchronization diagnostics

During real-hardware validation, monitor at least:

- packetsOk
- crcErrors
- packetsMissed
- resyncs
- RSSI
- current channel/frequency
- lastRadioError

A stable-looking lock with many missed slots can indicate timing, RF offset, antenna, or hop-sequence problems.

Persistence and provisioning

Rain totals and the RF rain counter are persisted in NVS. Wi-Fi credentials are also stored in NVS and are not compiled into public source. First boot uses the temporary DavisGateway-XXXX AP at 192.168.4.1; DHCP is the default LAN mode, while 192.168.1.120 is only the suggested static address.

Known limits of Edition 1.0

1. The guide describes the 0.2.0-dev decoder, not a stable release.
2. Unlisted packet types are not decoded.
3. Bytes 8-9 are not validated in the current alpha.
4. UV, direction and some rain-rate details require field verification.

5. SX1276 synchronization is project logic, not an official Davis algorithm.

Public technical references

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